FEI HAN

Robot Learning · Dynamics Modeling · Bioinspired Robotics

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Joint PhD Program · Expected completion June 2027

Education

Westlake University and Zhejiang University
Joint PhD Program · Computer Science and Technology
Aug 2023 - Jun 2027 (expected) · Advisor: Prof. Dixia Fan · i4FSI Lab

University of Chinese Academy of Sciences · MEng · Sep 2019 - Jun 2022
Pattern Recognition and Intelligent Systems
Research at the State Key Laboratory of Robotics, Shenyang Institute of Automation, CAS.

Zhengzhou University · BEng
Measurement and Control Technology and Instruments · Sep 2015 - Jul 2019

Doctoral Research Project

Bioinspired Propulsion and Robot Learning

Dec 2023 - Present · Westlake University · Doctoral research

Conduct parallel research on the REEF propulsor platform and quadruped swimming, connecting force measurements, dynamics modeling, policy learning and robot experiments.

Experimental data and modeling. Collected and processed force data; developed an LSTM hydrodynamic model and combined it with multi-objective gait optimization and robot validation. Contributed to REEF platform development, synchronized acquisition and force-feedback experiments.

Learning and control. Contributed to safe reinforcement learning with imitation initialization, constrained policy optimization and leg-phase coordination. Studied offline learning and online adaptation, and transferred single-leg policies to whole-body swimming.

Data and planning. Study dynamic-load reconstruction datasets and benchmarks, alongside hybrid behavior planning and skill distillation in an experiment-calibrated swimming simulator.

Selected Project Outputs

Learn to Swim · ICRA 2025 · Published · First author

Swimming Under Constraints · ICRA 2026 · Accepted · Co-first and co-corresponding author

REEF / Embracing Flow Unsteadiness · Nature Mechanical Engineering · Under review · Co-first author, listed first

From Gaits to Routes · ICRA 2027 · Under review

HydroWrench · ICLR 2027 · Under review

Applied Collaboration

DEEP Robotics Arctic quadruped. Contributed a force-testing, modeling and motion-parameter-optimization framework adopted by the collaboration, and joined specialized foot-design discussions. Application: Arctic expedition, Aug 2025.

Research and Engineering Experience

Multilegged Swimming and Bioinspired Locomotion

Apr 2026 - Present · Ongoing collaboration · Manuscript in preparation

Visiting Scholar at Pennsylvania State University, Apr - Oct 2026. Develop reduced-order dynamics models and numerical simulations for multilegged swimming, and contribute to experiment design.

Collaborate with researchers from Harvard University, Carnegie Mellon University and other institutions; the manuscript is in preparation. | Baxi Chong group

Storm-Petrel Locomotion

Jul 2022 - Oct 2023 · Westlake University

Combined air-water modeling with deep reinforcement learning to study storm-petrel pattering and sea-anchoring. Co-first-author paper in Bioinspiration & Biomimetics, 2023. Research Assistant: Jul 2022 - Aug 2023; continued into early PhD study.

Deep-Sea Soft Robotic Gripper

Sep 2019 - Jun 2022 · MEng research · Core student contributor

Contributed to NSFC Major Research Plan Integrated Project No. 92048302: Fundamental Research on Robots Combining Rigid, Flexible and Soft Structures for Operations in Extreme Environments.

Led structural design, fabrication, sensing and control development. Discussed technical approaches with the advisor and guided junior students in implementation and experiments; their later projects built on the structure and sensing work. Contributed to background research and preparation of project proposal materials.

Zhengzhou University RoboMaster Team

Sep 2016 - 2019 · Captain: 2017-2018 academic year · Advisor: 2019 season

Participated in the 2017, 2018 and 2019 seasons, serving as an advisor in 2019. Developed STM32-based hardware and C control software for infantry, hero, engineer, aerial and sentry robots, including chassis, gimbal and projectile-feeding controls.

As captain, coordinated technical planning, task allocation, subsystem integration and development schedules while remaining involved in implementation and troubleshooting.

Methods and Tools

Learning and modeling: temporal models, constrained RL, imitation learning, policy distillation and multi-objective optimization.
Implementation: Python, MATLAB, MuJoCo, ROS, STM32/C; force sensing, synchronized acquisition and robot integration.